Computation of Conformal Mappings

Report submitted to

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In partial fulfilment of the requirements for the award of the degree of

B.S.(Hons.) in Mathematics and Computing

By

Ravi Kumar

22MA10047

Under the guidance of

Prof. Bappaditya Bhowmik

DEPARTMENT OF MATHEMATICS
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

November 2025

DECLARATION

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Prof. Bappaditya Bhowmik
Department of Mathematics
Supervisor

Date: November 2025

Abstract

This report provides a comprehensive analysis of the **Schwarz-Christoffel (SC) Conformal Mapping**, a fundamental tool in complex analysis for transforming the upper half-plane onto the interior of any simple polygon. It begins by establishing the theoretical foundations of conformal mapping, including the **Riemann Mapping Theorem** and essential properties of analytic functions. The report then develops the derivation and application of the SC formula, explaining how the differential expression relates to the exterior angles of the target polygon and how the mapping is constructed. A detailed mathematical discussion is provided, along with illustrative examples involving elliptic integrals and mappings of basic polygonal regions. The report concludes by highlighting the practical significance of SC mappings in fields such as fluid dynamics, heat transfer, and grid generation, supported by clear visual explanations and diagrams to enhance understanding.

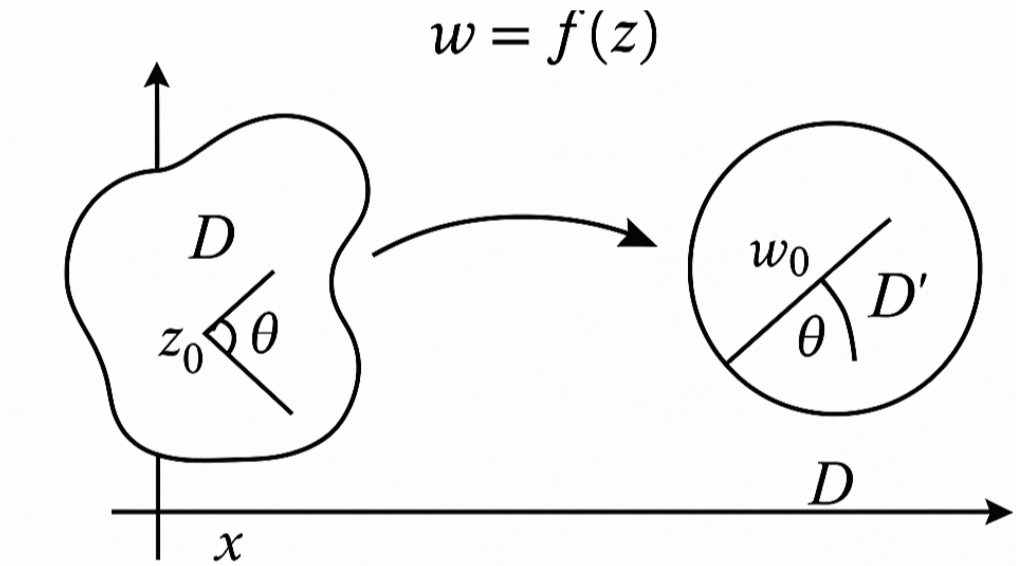


Figure 1: Schwarz-Christoffel transformation overview

Contents

Abstract **1**

1 Introduction to Conformal Mapping **3**
1.1 Historical Context and Foundations 4
1.2 Analytic Functions and Conformality 4
1.2.1 Properties of Conformal Mapping 5

2 Foundational Theorems of Conformal Mapping **6**
2.1 The Riemann Mapping Theorem 6
2.2 The Bilinear (Möbius) Transformation 7

3 The Schwarz-Christoffel Transformation **9**
3.1 Purpose and General Formula 9
3.2 Derivation Concept: The Argument Principle 10
3.2.1 Step-by-Step Derivation Concept 11
3.2.2 Construction Steps 11
3.3 Parameters of the Transformation 12

4 Examples of SC Mapping **13**
4.1 Mapping to a Rectangle and Elliptic Integrals 13
4.2 Mapping to a Semi-Infinite Strip 14
4.3 Mapping to a Triangular Region 14

5 Applications of Conformal Mapping **16**
5.1 Potential Theory and Boundary Value Problems 16
5.2 Fluid Dynamics and Aerodynamics 16
5.3 Engineering and Numerical Methods 17

6 Conclusion **19**

List of Important Symbols **20**

References **20**

1 Introduction to Conformal Mapping

The study of conformal mapping is a cornerstone of complex analysis, providing a powerful mathematical framework for solving boundary value problems in two dimensions. A conformal map is a function that preserves the angles between intersecting curves, both in magnitude and sense (orientation). This property is invaluable in transforming complex geometries into simpler, standard domains where solutions to physical problems, such as those governed by Laplace's equation, are readily available. The ability to transform a complex physical domain into a simple computational domain, solve the problem there, and then map the solution back to the physical domain is the central utility of conformal mapping. This technique is not just a mathematical curiosity but a practical tool that has shaped the fields of fluid dynamics, electrostatics, and computational geometry. The preservation of the governing equations (like Laplace's equation) under conformal transformation is the key mathematical property that makes this approach so effective. This report will focus on the most explicit and constructive tool in this domain: **the Schwarz-Christoffel Transformation**.

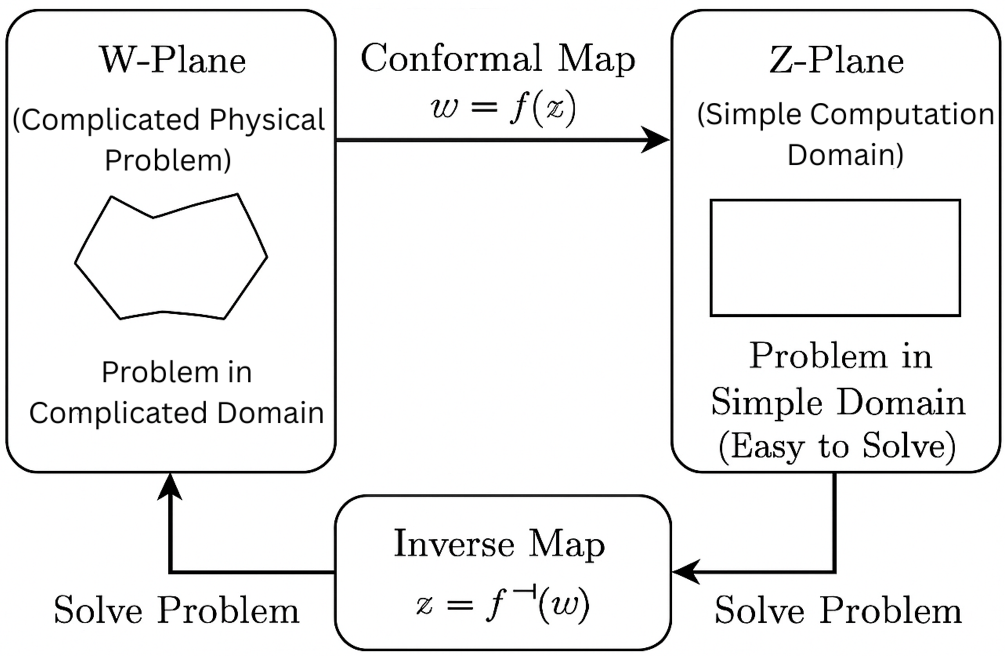


Figure 2: Transformation illustration

The ability of conformal maps to preserve local shape, while allowing for global distortion, is what makes them so powerful. This preservation of angles is mathematically equivalent to the mapping function being analytic, a condition that forms the bedrock of complex analysis and its applications to physical problems. The transformation effectively allows engineers and physicists to solve complex boundary value problems in a simplified domain and then map the solution back to the original, intricate geometry.

1.1 Historical Context and Foundations

The concept of angle-preserving transformations has roots dating back to ancient cartography, notably with Claudius Ptolemy's stereographic projection and Gerhard Kremer's Mercator projection in 1569 . However, the rigorous mathematical foundation was established much later. Carl Friedrich Gauss first solved the general problem of determining all conformal mappings for a small neighborhood of a point in 1822 .

The major breakthrough came with **Bernhard Riemann** in 1851, who formulated the **Riemann Mapping Theorem**. While Riemann's initial proof relied on the assumption that the Dirichlet problem had a solution, the theorem was later rigorously established by **Hermann Schwarz** in 1890 and **David Hilbert** in 1901 . The historical development of computational methods, including the work of Lichtenstein (1917) and Theodorsen (1931), highlights the continuous effort to find practical ways to compute these complex transformations .

The first numerical research into developing a method for mapping a region bounded by finitely many Jordan curves was done by Burnside in 1891 . A minimizing principle was established by **Bieberbach** in 1914, namely, that among all suitably normed conformal maps of a given simply connected region the one with the least area is the conformal map onto a circle . The development of integral equation methods by **Lichtenstein** in 1917, who reduced the problem to the solution of an integral equation, marked a significant step forward . Other attempts in this direction were made by Krylov and Bogolyubov in 1929. A significant contribution was made by **Nyström** in 1930, who gave a method for an approximate solution of integral equations useful in conformal mapping . The important result known as **Theodorsen's integral equation** was developed in 1931 for conformally mapping nearly circular regions onto a circle, which further solidified the field's practical application . This historical trajectory highlights the continuous shift from purely theoretical existence proofs to practical, computational methods for constructing conformal maps, often involving the solution of complex integral equations.

1.2 Analytic Functions and Conformality

The mathematical condition for a transformation $w = f(z)$ to be conformal is intrinsically linked to the properties of **analytic functions**. A function $f(z)$ is analytic (or holomorphic) in a region D if its derivative $f'(z)$ exists at every point in D.

The necessary condition for analyticity is the **Cauchy-Riemann Equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

where $f(z) = u(x, y) + iv(x, y)$.

The connection to conformal mapping is direct:

- **Sufficient Condition:** If $f(z)$ is analytic in domain D and its derivative $f'(z) \neq 0$ for all $z \in D$, then $f(z)$ is a conformal mapping in D.
- **Local Univalence:** The condition $f'(z) \neq 0$ ensures that the mapping is locally one-to-one (injective), meaning small neighborhoods map to small neighborhoods without overlap.
- **Harmonic Functions:** The real and imaginary parts of an analytic function, u and v, are **harmonic functions** (i.e., they satisfy Laplace's equation, $\nabla^2 u = 0$ and $\nabla^2 v = 0$). This property is crucial because it allows boundary value problems defined by Laplace's equation to be solved in the simpler transformed domain.

1.2.1 Properties of Conformal Mapping

A key property of conformal maps is the preservation of orthogonality, which is a direct consequence of the Cauchy-Riemann equations. The level curves of $u(x, y) = \text{constant}$ and $v(x, y) = \text{constant}$ are orthogonal at every point where $f'(z) \neq 0$.

Local Scale Factor: While angles are preserved, distances are not. However, the ratio of infinitesimal distances is preserved, which is defined by the **local scale factor** or **magnification factor**, λ:

$$\lambda = |f'(z)| = \left| \frac{dw}{dz} \right|$$

This factor is independent of the direction of the curve at z. If ds_w is an infinitesimal length in the w-plane and ds_z is the corresponding length in the z-plane, then $ds_w = \lambda ds_z$. This means that a small circle centered at z is mapped to a small circle centered at $w = f(z)$, but with a radius scaled by λ.

Preservation of Orientation: Conformal maps also preserve the orientation (sense) of the angle. A mapping that preserves the magnitude of angles but reverses the orientation is called an **isogonal** mapping. For an analytic function $f(z)$, the derivative $f'(z)$ is a complex number whose argument, $\arg(f'(z))$, represents the angle of rotation of the tangent vector at z.

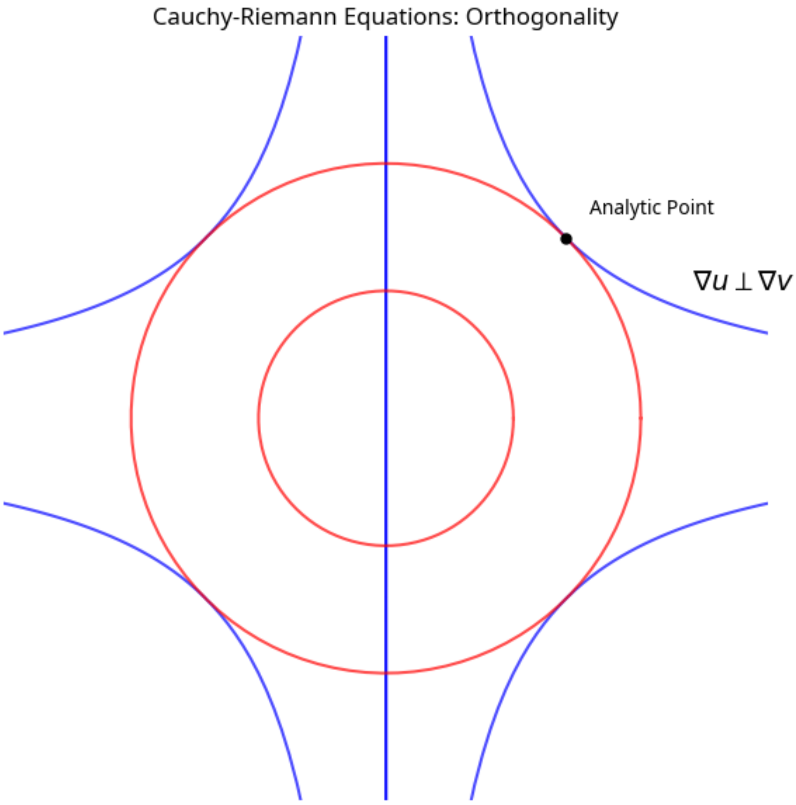


Figure 3: Cauchy-Riemann Orthogonality

2 Foundational Theorems of Conformal Mapping

2.1 The Riemann Mapping Theorem

The **Riemann Mapping Theorem** is the foundational result that guarantees the existence of a conformal map for a vast class of domains.

Theorem (Riemann Mapping Theorem): Let $D \subset \mathbb{C}$ be a simply connected region, and $D \neq \mathbb{C}$. Then there exists a conformal mapping $f : D \to U$ that maps D onto the open unit disk $U = \{w \in \mathbb{C} : |w| < 1\}$.

The theorem has profound significance:

- **Existence:** It guarantees that a conformal mapping from any simply connected domain (excluding the entire complex plane) to the unit disk always exists.
- **Uniqueness:** The mapping is unique up to normalization (e.g., fixing the image of a point and the argument of the derivative at that point).
- **Significance:** This theorem enables the reduction of complex boundary value problems defined on arbitrary simply connected domains to standard problems on the unit disk, where solutions are well-known.

The theorem essentially states that all simply connected domains, no matter how complex their boundary, are conformally equivalent to the simplest possible domain, the unit disk. The uniqueness of the mapping is a critical aspect, ensuring that once the normalization conditions are fixed (e.g., fixing the image of a point and the argument of the derivative at that point), the conformal map is uniquely determined. This property is vital for applications, as it guarantees a consistent and predictable transformation for solving boundary value problems. The rigorous proof of the theorem, completed by Schwarz and Hilbert, confirmed the foundational role of this concept in complex analysis.

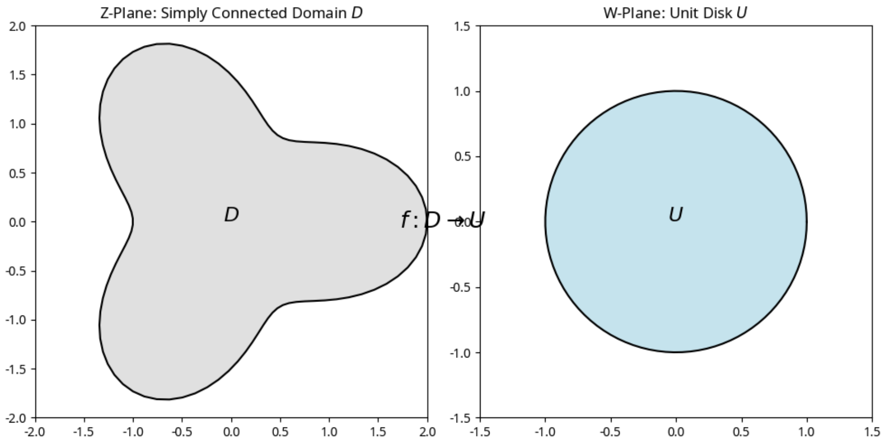


Figure 4: Riemann Mapping Theorem Concept

2.2 The Bilinear (Möbius) Transformation

A simple, yet powerful, class of conformal maps is the **Bilinear (Möbius) Transformation**, defined by:

$$w = \frac{az + b}{cz + d}, \quad \text{where } ad - bc \neq 0$$

This transformation is conformal everywhere except at the pole $z = -d/c$. Its key properties are:

- **Circle/Line Preservation:** It maps circles and straight lines in the z-plane to circles or straight lines in the w-plane. Straight lines are considered circles with infinite radius.
- **Composition:** Any bilinear transformation can be decomposed into a sequence of simpler transformations: translation, rotation/magnification, and inversion.
- **Normalization:** The condition $ad - bc \neq 0$ ensures the transformation is non-degenerate and invertible.

A common application is mapping the upper half-plane (UHP) to the unit disk, as illustrated below. The transformation $w = \frac{z - z_0}{z - \bar{z}_0}$ maps the UHP to the unit disk, where z_0 is a point in the UHP. The decomposition of the Bilinear Transformation is particularly insightful: it can be viewed as a sequence of elementary transformations:

1. **Translation:** $z_1 = z + d/c$

2. **Inversion:** $z_2 = 1/z_1$

3. **Rotation and Magnification:** $w = \frac{ad-bc}{c} z_2 + \frac{a}{c}$

This decomposition demonstrates the fundamental nature of the Bilinear Transformation, showing how complex mappings can be built from simple geometric operations. Its property of mapping circles and lines to circles and lines makes it an indispensable tool for simplifying domains in the initial stages of a conformal mapping problem.

Furthermore, the Bilinear Transformation is involutory when $w = \frac{az+b}{cz+d}$ and $z = \frac{dw-b}{-cw+a}$, which occurs when $a = -d$. This property is important in certain geometric applications where the transformation is its own inverse.

The Bilinear Transformation is often used as a preliminary step in more complex mappings, allowing for the standardization of one part of the domain before applying a more specialized transformation like the Schwarz-Christoffel formula.

Decomposition of the Bilinear Transformation The Bilinear Transformation, a fundamental conformal map, can be understood as a sequence of simpler geometric operations: translation, inversion, and rotation/magnification. This decomposition is crucial for understanding how complex maps are built from elementary transformations.

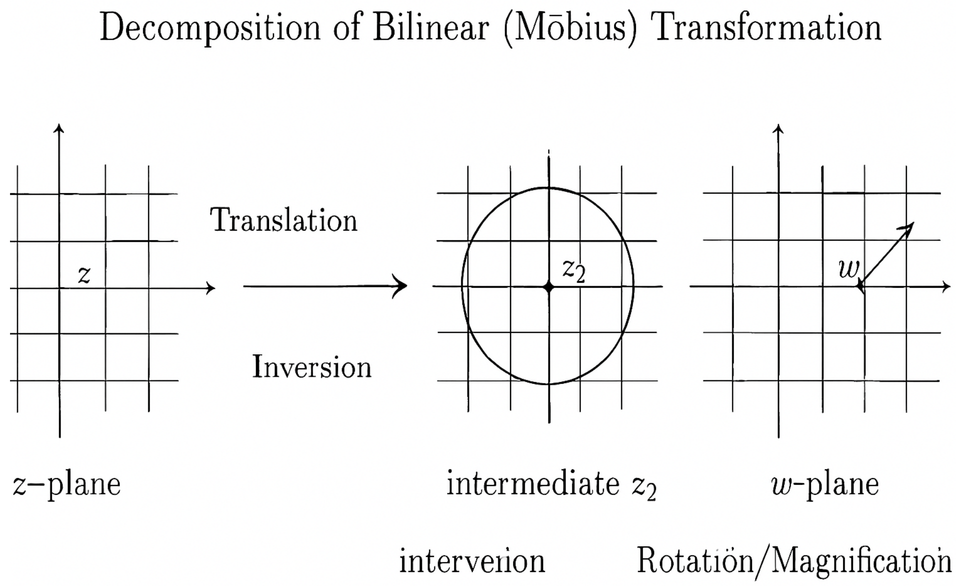


Figure 5: Bilinear Transformation Decomposition

3 The Schwarz-Christoffel Transformation

While the Riemann Mapping Theorem guarantees the existence of a conformal map, it does not provide an explicit formula for constructing it for an arbitrary domain. The **Schwarz-Christoffel (SC) Transformation** fills this gap for a specific, yet highly important, class of domains: simple polygons.

3.1 Purpose and General Formula

The SC transformation maps the upper half-plane (UHP), $\mathrm{Im}(z) > 0$, conformally onto the interior of any simple polygon with a finite number of vertices.

The differential form of the SC transformation is given by:

$$\frac{dw}{dz} = A \prod_{k=1}^{n} (z - x_k)^{-\alpha_k}$$

The mapping function $w = f(z)$ is obtained by integrating this expression:

$$w = A \int \prod_{k=1}^{n} (z - x_k)^{-\alpha_k} dz + B$$

Where:

- w is the complex coordinate in the polygon plane.
- z is the complex coordinate in the upper half-plane.
- x_k are n distinct points on the real axis of the z-plane ($x_1 < x_2 < \cdots < x_n$) that map to the vertices w_k of the polygon in the w-plane.
- $\alpha_k \pi$ is the **exterior angle** at the vertex w_k.
- A and B are complex constants determined by normalization conditions.

The interior angle θ_k of the polygon at vertex w_k is related to the exterior angle $\alpha_k \pi$ by:

$$\theta_k = (1 - \alpha_k)\pi \quad \text{or} \quad \alpha_k = 1 - \frac{\theta_k}{\pi}$$

For a closed polygon with n vertices, the sum of the exterior angles is 2π, which implies: $\sum_{k=1}^{n} \alpha_k = 2$.

The general concept of the mapping is illustrated below, showing the pre-images x_k on the real axis mapping to the vertices w_k of the polygon.

3.2 Derivation Concept: The Argument Principle

The Argument Principle: The core idea behind the SC formula is to control the change in the argument (angle) of the derivative $\frac{dw}{dz}$ as the point z moves along the real axis of the UHP.

Boundary Mapping: The real axis in the z-plane maps to the boundary of the polygon in the w-plane. As z traverses the real axis from left to right, the image point w traces the polygon boundary.

Vertex Control: At each vertex w_k, the path must turn by the exterior angle $\alpha_k \pi$.

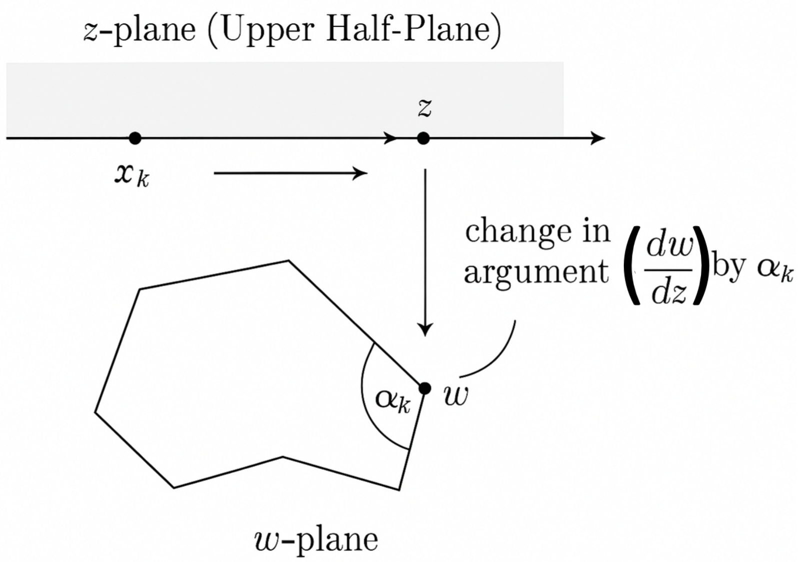


Figure 6: SC Derivation Concept

Schwarz-Christoffel Formula:

$$\frac{dw}{dz} = A \prod_{k=1}^{n} (z - x_k)^{-\alpha_k}$$

where A is a complex constant, x_k are the pre-images of the polygon vertices on the real axis, and α_k are the exterior angles.

Why $(z - x_k)^{-\alpha_k}$?
As z crosses x_k on the real axis (moving from $x_k - \epsilon$ to $x_k + \epsilon$ via the upper half-plane), the argument of $(z - x_k)$ changes by $-\pi$.
Therefore, the argument of $(z - x_k)^{-\alpha_k}$ changes by:

$$\Delta \arg((z - x_k)^{-\alpha_k}) = -\alpha_k \times (-\pi) = \alpha_k \pi$$

This is exactly the exterior angle turn required at vertex w_k.

Exterior Angle Relationship:

- Interior angle at vertex k: $\theta_k = (1 - \alpha_k)\pi$
- Exterior angle at vertex k: $\alpha_k \pi$
- Constraint: $\sum_{k=1}^{n} \alpha_k = 2$ (ensures polygon closes)

3.2.1 Step-by-Step Derivation Concept

1. **Start with the derivative:** We seek a function $w = f(z)$ that maps the upper half-plane conformally to a polygon. The derivative $\frac{dw}{dz}$ encodes how angles and shapes are transformed.

2. **Control the argument:** As z moves along the real axis, the argument of $\frac{dw}{dz}$ must change by the exterior angle $\alpha_k \pi$ at each vertex pre-image x_k.

3. **Construct the product form:** Each factor $(z - x_k)^{-\alpha_k}$ contributes the required argument change $\alpha_k \pi$ as z crosses x_k. The product of all such factors ensures simultaneous control at all vertices.

4. **Integrate to find w:** The mapping function is obtained by integrating:

$$w = A \int \prod_{k=1}^{n} (z - x_k)^{-\alpha_k} \, dz + B$$

where A and B are complex constants controlling scale, rotation, and translation.

3.2.2 Construction Steps

1. **Identify vertices:** Identify vertices $x_1, x_2, \ldots, x_n$ on the real axis that map to polygon vertices $w_1, w_2, \ldots, w_n$.

2. **Determine exterior angles:** Determine exterior angles $\alpha_k \pi$ from the interior angles θ_k of the polygon.

3. **Construct the product:** Construct the product $\prod_{k=1}^{n} (z - x_k)^{-\alpha_k}$ to ensure each factor contributes the correct argument change.

4. **Integrate to obtain w:** Integrate to obtain

$$w = A \int \prod_{k=1}^{n} (z - x_k)^{-\alpha_k} \, dz + B$$

where A and B are determined by normalization conditions.

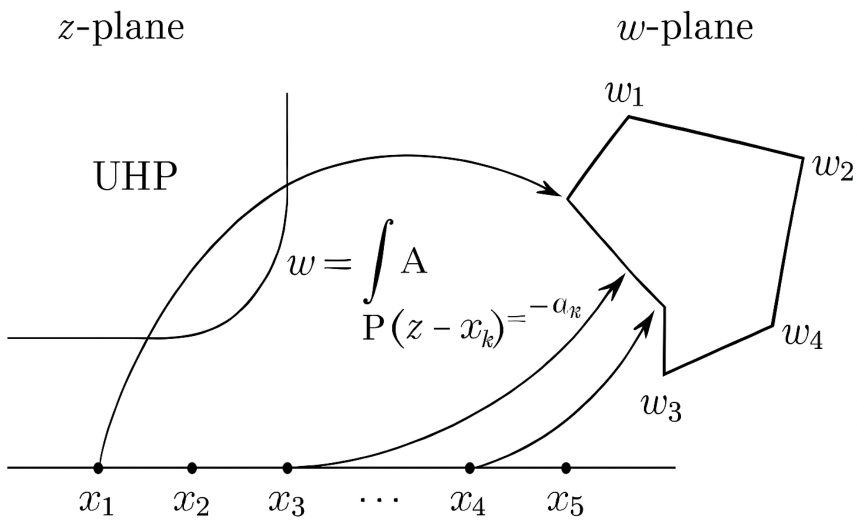


Figure 7: SC Formula Diagram

3.3 Parameters of the Transformation

The SC formula involves n pre-images x_k on the real axis, n exterior angles α_k, and two complex constants A and B.

The Location of Pre-images (x_k): The location of the n points x_k on the real axis determines the relative lengths of the sides of the polygon. For a polygon with n vertices, only $n-3$ of the pre-image points can be chosen arbitrarily. The remaining three points are fixed by the choice of the constants A and B, which control the scale, rotation, and translation of the resulting polygon. This freedom of choice is a consequence of the fact that a bilinear transformation can map any three points on the real axis to any other three points on the real axis, and the composition of a conformal map with a bilinear map is still a conformal map.

The Constants A and B:

- **Constant A (Scale and Rotation):** The magnitude $|A|$ determines the size (scale) of the polygon, and the argument $\arg(A)$ determines its orientation (rotation) in the w-plane.
- **Constant B (Translation):** The constant B determines the position (translation) of the polygon in the w-plane.

In practical applications, the values of x_k, A, and B are chosen to match the desired dimensions and location of the target polygon. This often involves solving a system of non-linear equations, which can be computationally intensive for complex polygons.

The Parameter Problem in Schwarz-Christoffel Mapping The parameter problem is the challenge of determining the pre-image points x_k on the real axis that correspond to the desired side lengths of the target polygon. This is a non-linear problem central to the practical application of the SC formula.

4 Examples of SC Mapping

The power of the Schwarz-Christoffel transformation lies in its ability to provide an explicit, albeit sometimes complex, integral form for the mapping function of various polygonal regions.

4.1 Mapping to a Rectangle and Elliptic Integrals

Mapping the UHP onto a rectangle is a classic example that demonstrates the complexity of the SC integral. A rectangle has four vertices, each with an interior angle of $\theta_k = \pi/2$.

The exterior angle is $\alpha_k \pi$, where

$$\alpha_k = 1 - \frac{\pi/2}{\pi} = 1 - \frac{1}{2} = \frac{1}{2}$$

Since there are four vertices, $n = 4$, and all $\alpha_k = 1/2$. The differential equation becomes:

$$\frac{dw}{dz} = A(z-x_1)^{-1/2}(z-x_2)^{-1/2}(z-x_3)^{-1/2}(z-x_4)^{-1/2} = \frac{A}{\sqrt{(z-x_1)(z-x_2)(z-x_3)(z-x_4)}}$$

The mapping function is:

$$w = A \int \frac{dz}{\sqrt{(z-x_1)(z-x_2)(z-x_3)(z-x_4)}} + B$$

The integral of this form is known as an **elliptic integral**. This demonstrates that even for a simple polygon like a rectangle, the mapping function cannot be expressed in terms of elementary functions, highlighting the need for numerical methods in practical SC applications. The ratio of the sides of the rectangle is determined by the ratio of two elliptic integrals, which is a key result in the theory of elliptic functions.

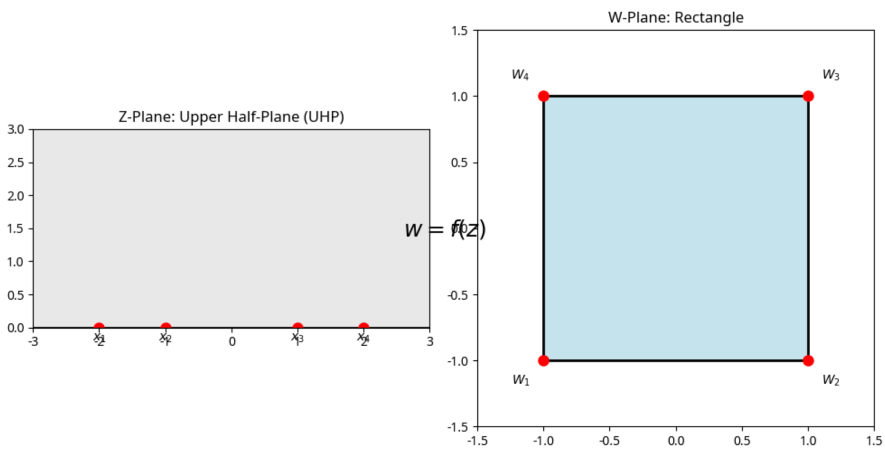


Figure 8: Rectangular Mapping Concept

4.2 Mapping to a Semi-Infinite Strip

A semi-infinite strip is a polygon with three vertices, two at a finite distance and one at infinity. Consider a strip defined by $0 < \mathrm{Im}(w) < h$ and $\mathrm{Re}(w) > 0$. The vertices are $w_1 = 0$, $w_2 = ih$, and $w_3 = \infty$.

The interior angles are $\theta_1 = \pi/2$ and $\theta_2 = \pi/2$. The exterior angles are $\alpha_1\pi$ with $\alpha_1 = 1/2$, and $\alpha_2\pi$ with $\alpha_2 = 1/2$. Since one vertex is at infinity, we only need two finite pre-images, x_1 and x_2. Let $x_1 = -1$ and $x_2 = 1$.

The differential equation is:

$$\frac{dw}{dz} = A(z - (-1))^{-1/2}(z-1)^{-1/2} = \frac{A}{\sqrt{z^2-1}}$$

The mapping function is:

$$w = A\int \frac{dz}{\sqrt{z^2-1}} + B = A\cosh^{-1}(z) + B$$

By choosing $A = h/\pi$ and $B = ih/2$, the mapping becomes:

$$w = \frac{h}{\pi}\cosh^{-1}(z) + \frac{ih}{2}$$

This maps the UHP to the semi-infinite strip. This is a case where the integral can be solved in terms of elementary functions, providing a closed-form solution.

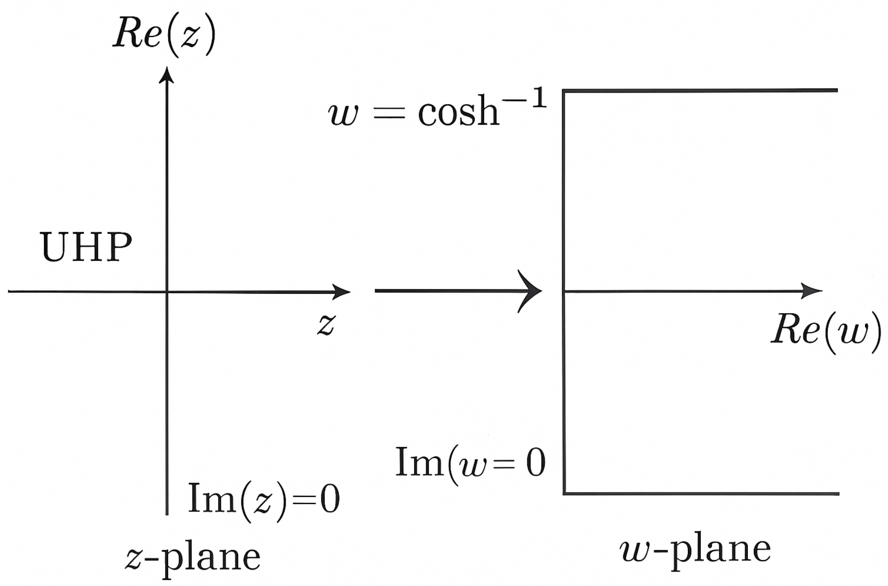


Figure 9: Semi-Infinite Strip Mapping Concept

4.3 Mapping to a Triangular Region

A triangle is the simplest closed polygon, with three vertices w_1, w_2, w_3 and interior angles $\theta_1, \theta_2, \theta_3$. The exterior angles are $\alpha_k = 1 - \theta_k/\pi$. Since $\sum \theta_k = \pi$, the sum of the exterior angles is $\sum \alpha_k = 3 - \sum \theta_k/\pi = 3 - 1 = 2$, which satisfies the closure condition.

The SC formula for a triangle with finite pre-images x_1, x_2, x_3 is:

$$w = A \int (z - x_1)^{-\alpha_1} (z - x_2)^{-\alpha_2} (z - x_3)^{-\alpha_3} dz + B$$

This integral is generally an **hypergeometric function** and cannot be solved in terms of elementary functions unless the angles are specific rational multiples of π.

A common simplification is to map one vertex to infinity, say $w_3 \to \infty$. This eliminates the factor $(z - x_3)^{-\alpha_3}$ from the differential equation, leaving only two finite pre-images x_1 and x_2. The mapping becomes:

$$w = A \int (z - x_1)^{-\alpha_1} (z - x_2)^{-\alpha_2} dz + B$$

This form is often easier to handle numerically or analytically for specific angle combinations.

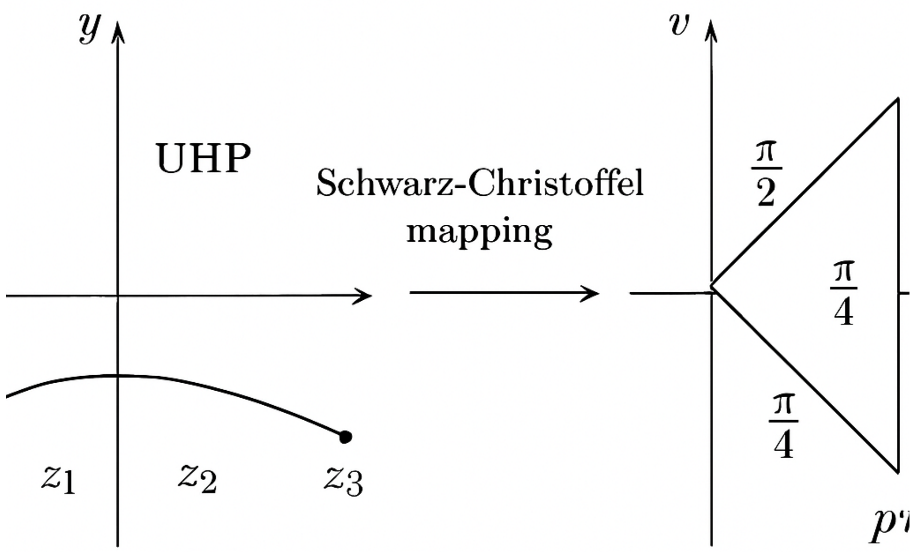


Figure 10: SC Triangle Mapping Concept

5 Applications of Conformal Mapping

Conformal mapping serves as a powerful mathematical tool for solving complex problems by transforming intricate domains into simpler, more tractable ones. This approach has proven particularly valuable in reducing challenging boundary value problems to standard forms that admit analytical or efficient numerical solutions.

5.1 Potential Theory and Boundary Value Problems

Conformal mapping provides elegant solutions to Dirichlet and Neumann problems in two dimensions, which are governed by Laplace's equation $\nabla^2 u = 0$. The technique enables the transformation of complex geometries into canonical domains where boundary value problems can be solved more readily.

- **Neumann Problem for Laplace's Equation:** Conformal mapping techniques have been successfully applied to solve the Neumann problem for regions exterior to circles, including the determination of Green's functions for such domains. This approach transforms the complex exterior region into a simpler computational domain.
- **Electrostatic Characterization:** The method of minimizing functionals, closely related to the Plateau problem, provides an electrostatic characterization that serves as an effective approach for determining conformal maps. This connection between variational methods and electrostatics offers both theoretical insight and computational advantages.

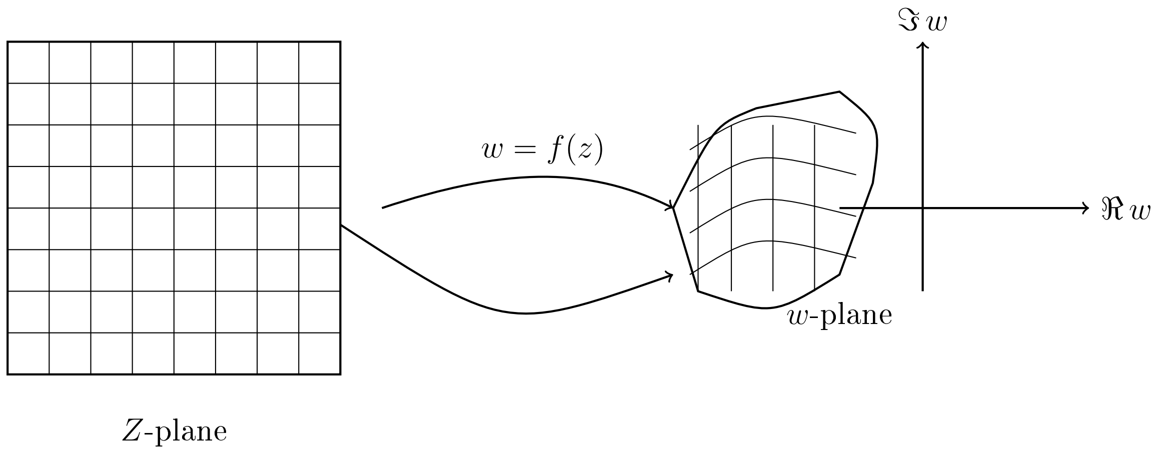
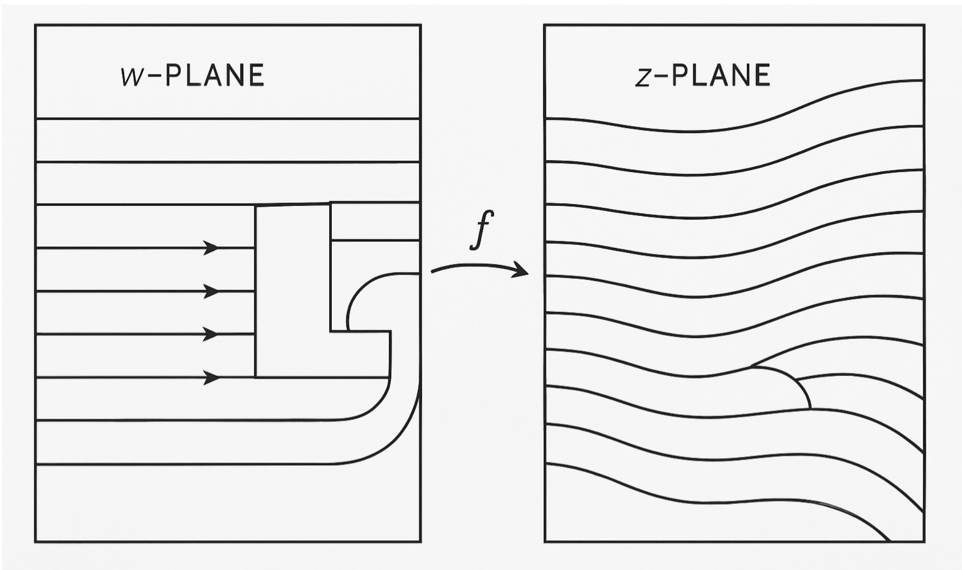


Figure 11: Conformal mapping from a rectangular grid in the z-plane to an irregular domain in the w-plane.

5.2 Fluid Dynamics and Aerodynamics

Conformal mapping has extensive applications in fluid mechanics, particularly for analyzing incompressible, irrotational flow problems where complex potential theory applies.

- **Airfoil Analysis:** The method of small parameters combined with conformal mapping enables the analysis of dynamic problems involving airfoils. This approach allows engineers to study aerodynamic properties and performance characteristics efficiently.
- **Flow and Heat Transfer in Conduits:** Conformal mapping facilitates the solution of flow and heat transfer problems in rectangular channels and layered geometries. By transforming complex channel cross-sections into simpler domains, the governing equations become more amenable to analytical or numerical solution.
- **Ideal Fluid Instability:** The technique has been applied to study the Rayleigh-Taylor instability in ideal fluids, where the conformal transformation simplifies the analysis of interfacial dynamics and stability characteristics.
- **Kirchhoff Flow:** Conformal mapping provides solutions to the classical Kirchhoff flow problem involving flow past polygonal obstacles, enabling the determination of velocity fields and pressure distributions around sharp-cornered bodies.



Application in Fluid Dynamics

Figure 12: Application of Conformal Mapping in Fluid Dynamics and Aerodynamics

5.3 Engineering and Numerical Methods

Modern engineering applications leverage conformal mapping for domain simplification in various computational and analytical contexts.

- **Thermoelastic Problems:** Conformal mapping assists in solving thermoelastic problems involving uniform heat flow perturbed by isolated holes of complex geometries, such as ovaloid forms. This approach enables the analysis of thermal stress distributions in components with geometric irregularities.
- **Stress Analysis in Propellant Grains:** The technique is employed to determine stress distributions in solid propellant rocket grains through the solution of integral equations. Mapping complex grain geometries to simpler domains facilitates both analytical and numerical stress analysis.

- **Acoustic Waveguides:** Conformal mapping enables the analysis of acoustic propagation in waveguides with complicated cross-sectional geometries, transforming them into canonical shapes where modal solutions are more readily obtained.
- **Rocket Motor Grain Design:** Boundary value and eigenvalue problems for solid propellant rocket motor grains are efficiently solved by conformally mapping the grain cross-section onto circles or annuli, where analytical solutions or specialized numerical methods are available.
- **Computational Grid Generation:** Conformal mapping has become an essential tool in computational engineering for generating boundary-fitted orthogonal grids. This application significantly enhances the accuracy and efficiency of numerical simulations by ensuring proper resolution near complex boundaries while maintaining grid quality.

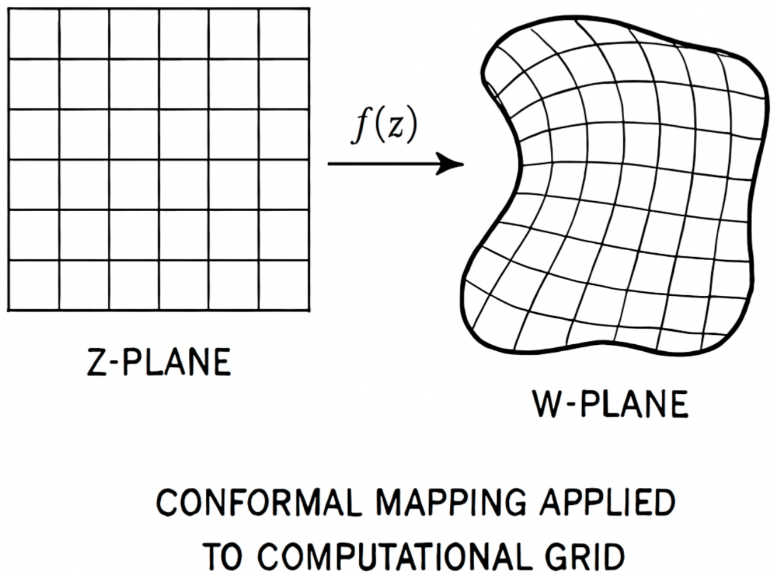


Figure 13: Boundary-Fitted Orthogonal Grid Generation Using Conformal Mapping

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6 Conclusion

Conformal mapping, rooted in the fundamental **Riemann Mapping Theorem**, provides an elegant and powerful mathematical framework for solving two-dimensional boundary value problems. The **Schwarz-Christoffel Transformation** is the explicit, constructive tool within this framework, allowing for the conformal mapping of the upper half-plane onto any simple polygonal region.

While the resulting SC integral is often complex, involving special functions like elliptic integrals, the formula's existence and structure provide a direct link between the geometry of the target polygon (via its exterior angles $\alpha_k \pi$) and the parameters of the mapping (the pre-images x_k). The practical utility of the SC transformation is demonstrated across diverse fields, from calculating lift on airfoils in **aerodynamics** to solving for potential fields in **electrostatics** and generating high-quality computational grids.

The interplay between theory and computation continues to drive innovation in this field. The ongoing development of robust numerical algorithms to solve the parameter problem and evaluate the resulting elliptic integrals ensures that the Schwarz-Christoffel transformation will continue to be a vital technique for modeling and solving real-world problems. The success of the Schwarz-Christoffel transformation is a powerful example of how pure mathematical theory can yield highly practical and indispensable tools for solving real-world problems. The continued research into numerical methods for the SC integral, particularly for complex and multiply-connected domains, ensures that this classical result will remain at the forefront of computational science for decades to come.

List of Important Symbols

- $z = x + iy$ - Complex variable in the source plane
- $w = u + iv$ - Complex variable in the target plane
- $f(z)$ or $w = f(z)$ - Conformal mapping function
- $f'(z)$ or $\frac{dw}{dz}$ - Derivative of the mapping function
- $\mathbb{C}$ - The complex plane
- UHP - Upper half-plane, $\{z \in \mathbb{C} : \text{Im}(z) > 0\}$
- $u(x, y)$, $v(x, y)$ - Real and imaginary parts of analytic function $f(z) = u + iv$
- $\nabla^2 u$ - Laplacian operator
- n - Number of vertices of the polygon
- w_k - The k-th vertex of the polygon in the w-plane
- x_k - Pre-image of vertex w_k on the real axis in the z-plane
- θ_k - Interior angle at vertex w_k
- α_k - Exterior angle parameter, where $\alpha_k = 1 - \frac{\theta_k}{\pi}$
- A, B - Complex constants in SC formula (scale, rotation, translation)

References

- **Driscoll, T. A., & Trefethen, L. N.** (2002). *Numerical Conformal Mapping: Domain Decomposition and the Mapping of Quadrilaterals.* World Scientific Publishing.